buildings

MDPI

Review

A Bibliometric Analysis of the Impact of Artificial Intelligence on the Development of Glass Fibre Reinforced Polymer Bars

Hajar Zouagho [1,*], Omar Dadah [1,2] and Issam Aalil [2]

[1] Laboratory of Intelligent Systems Energy and Sustainable Development, Private University of Fez, Fez 30999, Morocco; dadah@upf.ac.ma
[2] National Higher School of Arts and Crafts (ENSAM—Meknes), P.O. Box 15290, Meknes 50500, Morocco; i.aalil@umi.ac.ma
* Correspondence: hajar.zouagho@gmail.com

Abstract

Artificial Intelligence (AI) is increasingly shaping materials research, particularly in the development and optimization of Glass Fibre Reinforced Polymer (GFRP) bars used as innovative alternatives to steel reinforcement. Despite this growing intersection, no prior bibliometric study has systematically mapped how AI contributes to the advancement of GFRP technologies. This paper fills this gap through a comprehensive bibliometric analysis based on 102 Scopus-indexed publications from 2015 to 2025. Following PRISMA guidelines, the study combines performance analysis and science mapping using VOSviewer to identify publication dynamics, leading journals, key contributors, and thematic clusters. The results reveal a tenfold growth in annual output (compound annual growth rate, CAGR = 10.1%) and five dominant research directions: (1) machine learning in structural analysis, (2) AI-driven composite materials modeling, (3) smart damage detection, (4) mechanical characterization, and (5) advanced deep learning frameworks. China, India, and the United States collectively account for more than half of global publications, highlighting strong international collaboration. The findings demonstrate that AI has evolved from an exploratory tool to a transformative driver of innovation in GFRP research. This study provides the first quantitative overview of this emerging field, identifies critical gaps such as sustainability integration and standardization, and proposes future directions to foster cross-disciplinary collaboration toward intelligent and sustainable composite structures.

Keywords: artificial intelligence; machine learning; bibliometric analysis; glass fibre reinforced polymer bars; composite materials; sustainability; civil engineering applications; research trends

Check for updates

Academic Editors: Mohammad Al-Zu'bi, Rabee Shamass, Ahmed M. Ashteyat and Musab Rabi

Received: 2 September 2025
Revised: 3 November 2025
Accepted: 9 November 2025
Published: 28 January 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

Artificial Intelligence (AI) has emerged as a powerful driver of innovation across civil and structural engineering, enabling data-driven prediction, optimization, and decision support in materials and structures [1–3].

Within this context, Glass Fibre Reinforced Polymer (GFRP) bars have gained increasing attention as corrosion-resistant reinforcement for concrete and other cementitious systems, offering a favorable strength-to-weight ratio and high durability in aggressive environments [4–6].

Recent studies have demonstrated the potential of GFRP bars to enhance the service life of reinforced concrete structures, especially in marine or chemically aggressive conditions [7,8].

Their use aligns with sustainable-construction goals and circular-economy principles through lightweight design and corrosion mitigation [9,10].

As AI techniques mature from classical machine learning (ML) to deep learning (DL) and hybrid physics-informed models, their intersection with GFRP research is expanding, covering property prediction, design optimization, quality control, defect detection, and process monitoring [11–13].

For instance, AI-based models have been successfully used to predict bond strength, tensile properties, and fatigue behavior of FRP bars, as well as to automate quality-inspection procedures [14,15].

Despite this momentum, the knowledge landscape remains fragmented.

Research is dispersed across materials science, civil engineering, computer science, and manufacturing venues; terminology varies (e.g., "GFRP rebar" and "fibreglass reinforcement"), and datasets are heterogeneous.

The field, therefore, lacks a quantitative and reproducible mapping consolidating who is publishing, where, and on which methodological themes.

Several bibliometric or scoping reviews have addressed AI in construction or composite materials at large [16,17], but none focuses specifically on AI's impact on GFRP bar technologies' design, durability, manufacturing, inspection, and performance modeling over time and across regions.

A focused, methodologically transparent bibliometric analysis can reveal the structure of this fragmented literature, highlight evolutionary trends such as the post-2019 acceleration [6], and identify gaps, notably sustainability assessment and standardized validation to guide future research and industrial adoption [18,19].

This study systematically maps and quantifies how AI techniques have influenced the development of GFRP bars over the period 2006–2025.

We compile and analyze a curated corpus from Scopus (Elsevier B.V., Amsterdam, The Netherlands; https://www.scopus.com, accessed on 15 February 2025) and apply performance analysis and science-mapping to address three research questions:

RQ1. *How has scientific output on AI–GFRP evolved over time (venues, growth, geographic distribution)?*

RQ2. *Which topics and methods structure the field (keyword clusters, leading AI approaches, collaboration networks)?*

RQ3. *What gaps and opportunities emerge regarding design, durability/sustainability assessment, manufacturing, and standardization?*

This paper offers four main contributions:

1. First, a focused bibliometric mapping of AI–GFRP bars. We provide the first, to our knowledge, quantitative synthesis exclusively centered on AI applications to GFRP bar technologies [20,21].
2. Transparent and reproducible methodology. We report the exact Scopus advanced query, screening protocol with inter-rater reliability, cleaning/normalization rules, and a cleaned metadata file with DOIs [22,23].
3. Integrated insights into actors, methods, and themes. We identify leading countries, institutions, and authors; characterize thematic clusters and the adoption trajectory of AI techniques (ML, DL, hybrid) [11,12]; and relate them to venues and collaboration patterns.
4. Actionable research agenda. From the evidence, we articulate gaps (benchmark datasets, validation protocols, LCA and circularity integration, industrial deployment

barriers) and outline priorities for researchers and practitioners (standardized metrics, open datasets, domain-specific benchmarks) [9,10,13,19].

The analysis targets peer-reviewed articles and conference papers written in English and explicitly addressing AI in relation to GFRP bars (design, durability, manufacturing/QA, and monitoring).

Methodological details (database selection and justification, advanced query, screening and κ, normalization rules, thresholds, and clustering) are reported in Section 3.

Empirical trends and networks appear in Section 4 (Results), while implications, limitations, and future directions including sustainability and standardization are discussed in Section 5 (Discussion). Conclusions are presented concisely and numerically in Section 6.

2. Literature Review

2.1. Development and Applications of GFRP Bars

Over the past two decades, Glass Fibre Reinforced Polymer (GFRP) bars have been increasingly promoted as an alternative to conventional steel reinforcement in civil engineering. Their superior tensile strength, corrosion resistance, and low density make them attractive for applications in bridge decks, marine structures, and electromagnetically sensitive environments [24]. Despite these advantages, large-scale adoption remains constrained by concerns regarding long-term durability, variability in mechanical performance, and the absence of universally accepted design standards [25].

Recent advances have also expanded the scope of FRP applications beyond conventional reinforcement bars. Studies have examined the dynamic properties of thin-walled pultruded GFRP buckled columns using experimental and numerical investigations, highlighting the influence of local imperfections and cross-sectional stiffness on buckling behavior and modal response [26]. Other works have focused on enhancing the corrosion resistance of geopolymer concrete reinforced with basalt FRP (BFRP) bars in seawater environments, demonstrating superior durability and minimal degradation under long-term exposure [27]. These investigations reinforce the growing importance of FRP composites in sustainable infrastructure and underline how AI-driven modeling can further optimize their mechanical performance, corrosion resistance, and service-life prediction.

Beyond their structural performance, GFRP bars are increasingly used in research focusing on bond behavior, fatigue resistance, and hybrid reinforcement systems that combine FRP with steel or natural fibers [28,29]. Recent investigations have also addressed the integration of GFRP materials in smart monitoring systems, where embedded sensors and machine-learning algorithms enhance predictive maintenance and service-life assessment [30,31]. This progressive digitalization underscores the pivotal role of AI in enabling the next generation of adaptive and sustainable construction materials.

In addition to corrosion resistance and mechanical performance, the fire behavior of FRP reinforcement remains a critical safety concern for structural applications. Recent studies have shown that exposure to elevated temperature can significantly alter the tensile capacity and bond performance of FRP bars, and that thermal degradation mechanisms must be explicitly considered in design for serviceability and ultimate limit states [32]. These findings indicate that fire performance is still an underexplored dimension compared with corrosion and durability in marine environments, and they highlight the need for predictive models capable of assessing residual strength after a fire, potentially supported by AI-driven evaluation frameworks [32].

Additional recent studies have further addressed this issue by examining the fire resistance and post-fire mechanical behavior of FRP-reinforced concrete members [33]. These works emphasize the importance of integrating AI-based predictive models to improve

the assessment of GFRP performance under elevated temperature and fire exposure, thus addressing a critical research gap in this field.

Figure 1 illustrates typical Glass Fibre Reinforced Polymer (GFRP) bars used in civil engineering applications. Their ribbed surface and composite structure provide mechanical anchorage comparable to steel while ensuring superior corrosion resistance in aggressive environments.

Figure 1. Typical GFRP bars used as reinforcement elements in concrete structure.

2.2. The Rise in Artificial Intelligence in Composite Materials

Artificial Intelligence (AI) has recently emerged as a disruptive tool in the field of materials science. Machine learning and deep learning approaches enable the prediction of mechanical properties, detection of defects, and optimization of manufacturing processes [3]. Compared to traditional experimental procedures, these methods offer significant reductions in cost and time, while providing a deeper understanding of non-linear material behaviors [34]. In composite engineering, AI has been applied to improve design efficiency, enhance quality control in pultrusion and molding processes, and accelerate the development of high-performance structural components [1].

2.3. AI-Driven Innovation in GFRP Bars

Although still at an early stage, the integration of AI into GFRP bar research has already yielded promising outcomes. Neural networks and decision-tree models have been employed to predict residual tensile strength of bars exposed to alkaline environments [35]. Intelligent monitoring systems based on acoustic emission analysis have been proposed for damage detection and structural health monitoring [36]. Furthermore, the combination of AI with finite element modeling has opened new opportunities for optimizing composite structures and reducing design cycles [37]. These developments illustrate the growing convergence between civil engineering, computer science, and materials technology, where AI serves as a catalyst for accelerating innovation [38].

2.4. Circular Economy and Sustainability Dimensions

Beyond performance, sustainability considerations increasingly shape research on advanced construction materials. The circular economy framework emphasizes extending material lifespans, minimizing environmental footprints, and promoting recyclability [9]. Although GFRP production is often energy intensive, AI can contribute to sustainability by optimizing manufacturing parameters, reducing waste, and supporting life-cycle assessment of composites [10]. The integration of AI and circular economy principles has the potential to align technological innovation with global sustainability objectives, thereby enhancing the long-term viability of GFRP in infrastructure applications.

2.5. Bibliometric Studies and Research Positioning

In recent years, several bibliometric and scientometric studies have examined Artificial Intelligence (AI) applications across materials science and civil engineering domains [39,40]. For example, reviews on AI in composite materials have highlighted its use in predicting mechanical properties, detecting defects, and optimizing manufacturing parameters [39,41], whereas studies on AI in construction engineering emphasized process automation, BIM integration, and quality control [40]. However, none of these analyses specifically addressed glass-fiber reinforcement systems. This study therefore fills that gap by focusing on the intersection of AI, GFRP bar design, and structural performance.

Bibliometric methods have become an essential tool for mapping scientific knowledge and identifying emerging research domains. Recent works have applied bibliometric analyses to fields such as the bioeconomy, sustainable construction, and AI in engineering [16]. These studies highlight the value of systematic knowledge mapping to detect research gaps, track collaboration networks, and understand conceptual evolution [17]. However, despite the increasing attention given to AI in composite materials, no bibliometric study has specifically examined the intersection between AI and GFRP bar development. Addressing this gap is the primary aim of the present study, which offers, for the first time, a quantitative and critical overview of scientific production in this rapidly emerging field.

3. Methodology

3.1. Study Design

This quantitative bibliometric study systematically analyzes the scientific literature on the application of Artificial Intelligence (AI) to Glass Fibre Reinforced Polymer (GFRP) bars. The methodological framework follows the PRISMA 2020 guidelines adapted for bibliometric research [18]. The workflow was designed to ensure transparency, reproducibility, and alignment with the research questions presented in Section 1, in accordance with reviewers' recommendations.

The analysis integrates two complementary bibliometric approaches:

- Performance Analysis which quantifies publication and citation trends, research productivity, and geographic distribution [20].
- Science Mapping, which identifies intellectual structures, thematic clusters, and collaboration networks [20].

To maintain coherence with the research objectives, the study was structured around four research topics:

RT1. *Temporal Evolution—analyzed through publication and citation trends.*

RT2. *Key Stakeholders—examined via author, institutional, and country-level contributions.*

RT3. *Dominant AI Techniques and Applications—investigated through keyword co-occurrence and thematic clustering.*

RT4. *Research Gaps and Emerging Opportunities—explored through the identification of under-represented or emerging topics.*

The methodological design (Figure 2) is organized into five sequential stages, ensuring a systematic and replicable process:

- Database Selection (Section 3.2): justification of Scopus as the data source and formulation of the Boolean query;
- Data Retrieval (Section 3.2): execution of the advanced search and export of bibliographic records;

- Screening and Eligibility (Section 3.3): automatic and manual filtering following the PRISMA logic to retain relevant studies;
- Data Cleaning and Normalization (Section 3.3.3): harmonization of author names, institutional affiliations, and keywords; and
- Bibliometric Analysis and Visualization (Section 3.4): computation of performance indicators and mapping using VOSviewer 1.6.20 (Centre for Science and Technology Studies, Leiden University, Leiden, The Netherlands).

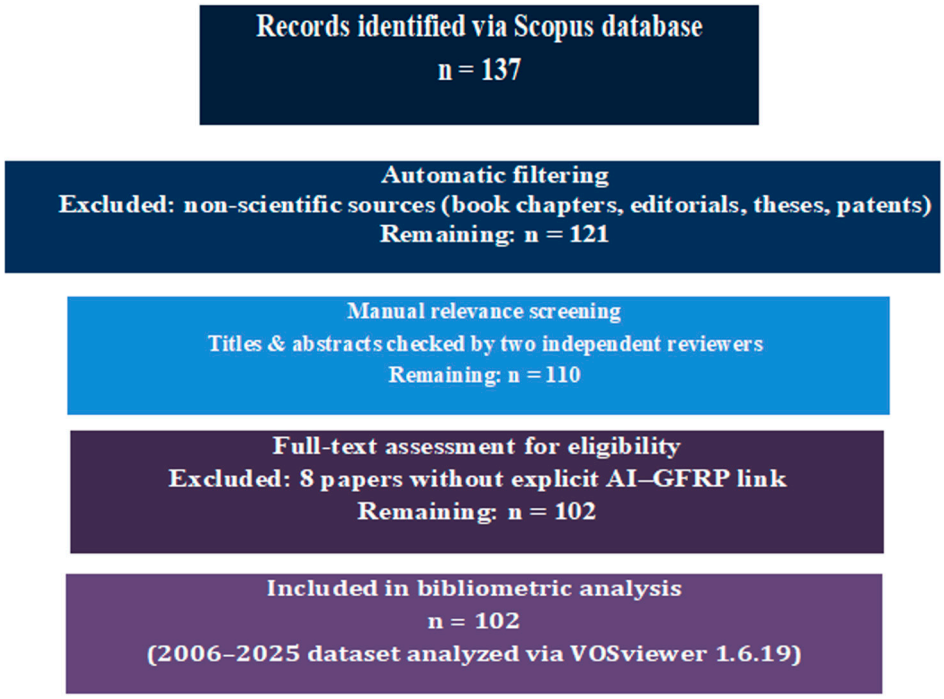

Figure 2. The PRISMA 2020 flow diagram illustrates the methodological workflow of the bibliometric study, highlighting the sequential screening and inclusion process.

This structured methodological framework guarantees that the collected data are reliable, the inclusion/exclusion process is traceable, and the results are reproducible, thereby fulfilling the reviewers' request for a clearer and more rigorous methodological scheme.

3.2. Database Selection and Search Strategy

The Scopus database was selected as the primary source because of its comprehensive coverage of engineering, materials science, and computer-science publications, ensuring a robust representation of AI-related studies on GFRP bars. Compared with the Web of Science (WoS), Scopus indexes a broader range of journals, conference proceedings, and open-access venues, providing richer metadata and interdisciplinary visibility [23,42]. Although WoS is recognized for its selectivity, its narrower indexing scope would have limited the coverage of emerging AI-driven research in construction and materials science.

The search was performed on 15 February 2025, and used the following Boolean expression, optimized through iterative testing for precision and recall:

TITLE-ABS-KEY ((“artificial intelligence” OR “AI” OR “machine learning” OR “deep learning”) AND (“glass fiber” OR “glass fibre” OR “GFRP”) AND (“bars” OR “rebar” OR “reinforcement”)) AND PUBYEAR > 2005 AND PUBYEAR < 2026 AND (EXCLUDE(SUBJAREA, ”AGRI”) OR EXCLUDE(SUBJAREA, “MEDI”) OR EXCLUDE(SUBJAREA, “NEUR”) OR EXCLUDE(SUBJAREA, “PSYC”))

Only peer-reviewed journal articles and conference papers written in English were retained.

Excluded items included non-scientific materials (book chapters, editorials, theses, and patents) and irrelevant domains such as medicine or neuroscience. The exported metadata contained authors, titles, abstracts, keywords, affiliations, and citation counts for subsequent screening and normalization (see Sections 3.3 and 3.4).

3.3. Screening Framework and Reliability Assessment

This study followed the PRISMA 2020 framework adapted for bibliometric research, ensuring a transparent and reproducible workflow from data retrieval to analysis [28].

The methodological process combined automatic screening, manual validation, and systematic normalization to guarantee the quality and accuracy of the dataset.

3.3.1. Screening Protocol and Reliability

The initial Scopus search retrieved 137 records (articles and conference papers). The screening was conducted in three phases:

1. Automatic filtering (137 → 121): exclusion of non-scientific records such as book chapters, editorials, theses, and patents.
2. Manual relevance screening (121 → 110): two independent reviewers examined titles and abstracts to confirm the presence of both AI and GFRP terms. Disagreements were resolved through discussion until consensus was reached.
   - Inter-rater agreement: $\kappa = 0.81$, indicating substantial reliability.
   - Each reviewer was trained on a random pilot subset ($n = 15$) to calibrate inclusion decisions.
   - Ties were adjudicated by a third co-author.
   - Confidence interval for κ: 0.74–0.87
3. Full-text assessment (110 → 102): exclusion of eight additional papers that lacked explicit connection between AI methods and GFRP bars.

The final corpus included 102 documents (2006–2025), representing a balanced dataset across engineering, materials, and computational disciplines.

3.3.2. Inclusion and Exclusion Criteria

Inclusion criteria

- Peer-reviewed journal articles or conference papers.
- Explicit focus on AI techniques applied to GFRP bars in design, durability, manufacturing, or monitoring.
- English-language publications within 2006–2025.

Exclusion criteria

- Studies unrelated to AI or not addressing GFRP reinforcement.
- Gray literature (reports, theses, patents).
- Duplicate or incomplete records.

3.3.3. Data Cleaning and Normalization

To ensure accurate bibliometric mapping, the dataset underwent a structured cleaning process using VOSviewer 1.6.20 and Microsoft Excel 2021 (Microsoft Corporation, Redmond, WA, USA):

- Author and Institution Standardization: harmonization of spelling variants and affiliation names (e.g., "Univ. of Toronto" → "University of Toronto").
- Keyword Thesaurus: a controlled vocabulary merged synonymous terms (e.g., AI, machine learning, artificial intelligence).
- Country Unification: merged sub-entities under national codes (e.g., "England," "Scotland," → "United Kingdom").
- Field Consistency: capitalization, duplicate removal, and correction of encoding errors.

3.4. Data Extraction and Bibliometric Analysis

Following the screening process described above, the bibliometric data were extracted from Scopus in CSV format and analyzed using a combination of quantitative indicators and network-based science-mapping techniques. Metadata fields (titles, authors, affiliations, keywords, sources, citations, and DOIs) were exported and cleaned as detailed in Section 3.3.

Two complementary bibliometric approaches were implemented to ensure a comprehensive and reproducible analysis [20,42]:

Performance Analysis: quantifies publication and citation trends over time, identifies the most productive authors, institutions, and countries, and highlights the most influential papers and journals.

Science Mapping: explores conceptual and collaborative structures of the field through co-occurrence, co-citation, and co-authorship networks, revealing thematic clusters and intellectual linkages among researchers.

3.4.1. Performance Indicators

Performance indicators were computed to describe temporal evolution and productivity patterns:

- Annual publication counts and citations were analyzed to detect growth phases and inflection points.
- Country and institutional productivity were normalized by total output to identify leading contributors.
- Highly cited documents and journals were ranked to determine intellectual influence within the corpus.

3.4.2. Science Mapping and Network Construction

Science mapping analyses were conducted using VOSviewer 1.6.20, a software developed by Van Eck and Waltman for constructing and visualizing bibliometric networks [43,44]. This tool enables the exploration of conceptual, intellectual, and social structures within the research field through the visualization of co-authorship, co-citation, and keyword co-occurrence networks.

To ensure a balance between precision and interpretability, the following parameters were adopted [31]:

- Keyword co-occurrence threshold: $\geq$3 occurrences.
- Author and source co-citation threshold: $\geq$5 citations.
- Clustering method: Leiden algorithm with default resolution = 1.0.
- Normalization method: association strength.

- Visualization type: network density map with weighted node sizing (frequency) and edge thickness (link strength).

All maps were manually inspected and refined to ensure semantic accuracy and visual coherence. This step prevented the inadvertent merging of distinct concepts (e.g., "deep learning" vs. "neural networks") and guaranteed consistent labeling across clusters.

The validated networks provided the foundation for the thematic and intellectual structure analyses presented in Section 4.

3.4.3. Reliability and Limitations

Methodological reliability was reinforced through dual validation of cleaning steps and a shared reference thesaurus. Nevertheless, common limitations inherent to bibliometric analyses apply, notably single-database dependence (Scopus), English-language bias, and citation-time lag [22]. These constraints were mitigated through methodological transparency, reproducible parameterization, and triangulation between performance and network-based insights.

Artificial Intelligence tools were used in this study. Specifically, ChatGPT (OpenAI, San Francisco, CA, USA; https://chat.openai.com, accessed on 2 November 2025) was employed to generate Figure 1 illustrating typical GFRP bars used as reinforcement elements in concrete structures. DeepL (DeepL SE, Cologne, Germany; https://www.deepl.com, accessed on 20 April 2025) Translator was used to translate parts of the manuscript from French to English.

4. Results: Bibliometric Analysis of the Impact of AI on the Development of GFRP

4.1. Publication by Year

Figure 3 illustrates the temporal evolution of publications addressing the application of Artificial Intelligence (AI) in the development and analysis of Glass Fibre Reinforced Polymer (GFRP) bars from 2006 to 2025. The scientific output remained modest and intermittent for over a decade, with only a few isolated studies published between 2006 and 2018. This long period of stagnation can be attributed to the limited availability of open-access datasets, the high cost of experimental campaigns on composite materials, and the absence of AI frameworks specifically adapted to structural and materials engineering during that time [11,45].

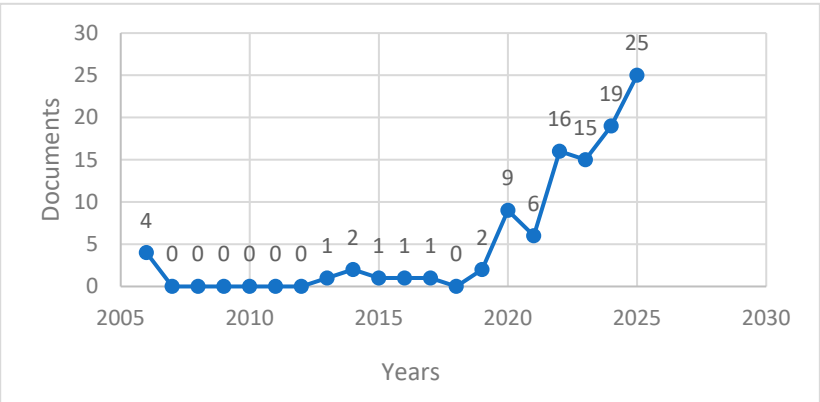

Figure 3. Annual evolution of publications on the application of Artificial Intelligence (AI) to GFRP bars (2006–2025).

A first inflection point appeared in 2019, when publication activity began to rise. This change reflects the growing accessibility of machine learning libraries and compu-

tational tools, along with the increasing recognition of AI's potential for optimizing the design, testing, and mechanical characterization of composite reinforcement bars [14,15]. The dissemination of open-source platforms and cloud-based computing environments also lowered technical barriers, enabling smaller research teams to engage in exploratory applications [12].

From 2020 onward, the field experienced a marked surge, with the annual number of publications increasing from 9 in 2020 to 25 in 2025. Several contextual factors contributed to this acceleration: the COVID-19 pandemic, which redirected many laboratories toward simulation- and data-driven research during periods of restricted experimental activity [19]; the emergence of high-performance and durability-optimized GFRP products that required advanced predictive and monitoring approaches [6]; and the broader inclusion of AI-based methodologies in civil engineering education and research programs [13].

This dynamic was further strengthened after 2022 through international funding schemes promoting digitalization, sustainability, and smart infrastructure [46], leading to a stabilized annual output exceeding 15 publications.

Over the entire 2006–2025 period, scientific production shows a compound annual growth rate (CAGR) of 10.1%, calculated as follows:

$$\mathrm{CAGR} = \left(\frac{N_{2025}}{N_{2006}}\right)^{\frac{1}{t}} - 1$$

where $N_{2025} = 25$, $N_{2006} = 4$, $t = 19$, confirming a steady and sustained expansion of research in this domain.

The annual publication counts used to calculate this CAGR are provided in Table S1 (Supplementary Materials). These data confirm the stagnation observed until 2018, followed by the sharp acceleration illustrated in Figure 3. A Poisson regression of annual publication counts versus year confirmed the statistical significance of this upward trend ($p < 0.05$), validating the robustness of the growth pattern.

Overall, this trajectory indicates that AI-based investigations on GFRP bars have evolved from an exploratory niche into a rapidly maturing and interdisciplinary field. These findings directly support RQ1 by evidencing the structural transformation of the field, driven by technological innovation, methodological democratization, and cross-disciplinary collaboration [47].

4.2. Journal Contributions

Figure 4 presents the distribution of publications across major journals and conference proceedings, illustrating how research on AI applications in GFRP bars has evolved from an exploratory stage to a phase of methodological consolidation within recognized scientific outlets.

A significant proportion of the studies (31%) were published in the IOP Conference Series: Materials Science and Engineering, reflecting the initially experimental and formative nature of this research area. Early investigations were frequently disseminated through conference proceedings to facilitate rapid feedback and scholarly interaction prior to submission to peer-reviewed journals [48]. This prevalence suggests that, during the early years, conferences served as incubators for testing data-driven frameworks and preliminary AI-based models before achieving methodological stabilization.

In contrast, three major peer-reviewed journals, Construction and Building Materials (22%), Journal of Building Engineering (17%), and Structures (17%) together account for more than half of all journal-based publications. Their strong representation underscores the growing maturity and credibility of AI–GFRP research, as studies published in these venues typically demonstrate robust experimental protocols and validated computational

approaches [49,50]. The increasing presence of GFRP–AI studies in these high-impact outlets indicates that this line of research is now perceived as scientifically rigorous rather than exploratory [51].

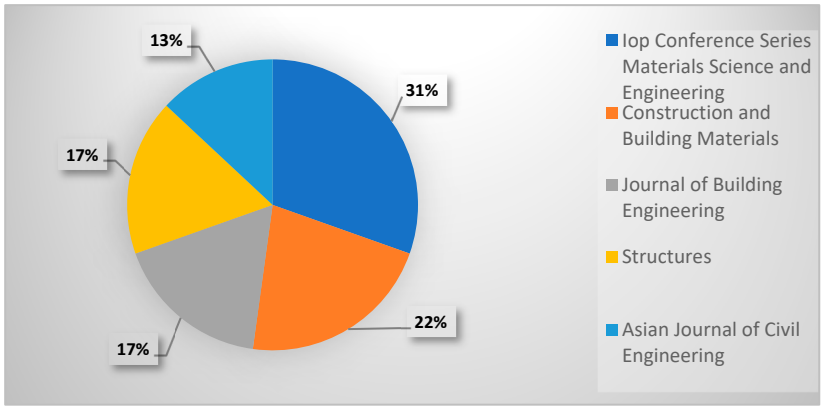

Figure 4. Distribution of publications across major journals and conference series (2006–2025). Data source: Scopus (query in Section 3.2).

The Asian Journal of Civil Engineering (13%) further highlights the increasing participation of research groups from Asia and the Middle East. This trend aligns with broader regional developments emphasizing infrastructure modernization and governmental investment in digital construction technologies [52]. Such geographic diversification not only broadens the empirical scope of the literature but also introduces new case studies, materials, and climatic contexts, enhancing the external validity of this research domain [53].

Overall, the concentration of publications within a limited number of high-impact journals and proceedings evidences a structural transition from a dispersed, exploratory phase toward a consolidated and recognized research discipline. The decreasing reliance on conference proceedings in favor of indexed, peer-reviewed journals signals a clear progression toward methodological standardization and scientific legitimacy [54].

4.3. Areas of Research into the Application of Artificial Intelligence in Fibreglass Bars

Figure 5 presents the disciplinary distribution of publications related to the application of Artificial Intelligence (AI) in Glass Fibre Reinforced Polymer (GFRP) bars. The results reveal a clear predominance of engineering (44%) and materials science (28%), followed by physics and astronomy (8%), computer science (7%), chemistry (4%), mathematics (4%), chemical engineering (3%), and environmental science (2%). This profile reflects both the applied orientation of the field and its gradual expansion toward interdisciplinary research domains.

To avoid over-representation of multidisciplinary papers, documents indexed under multiple Scopus subject areas were fractionally counted (1/n per category, where n is the number of assigned areas). This normalization ensures that the total number of records remains consistent with the retrieved dataset. A complementary validation using author-provided keywords confirmed similar proportional trends across the main disciplines (Engineering, Materials Science, and Computer Science), supporting the robustness of the subject-area analysis [55,56].

In addition, a manual validation on a random 20-paper subset, independently coded by two reviewers using author-provided keywords rather than Scopus categories, yielded an inter-coder agreement of 0.86 on top-level field assignment (Engineering/Materials Science/Computer Science), confirming the consistency of the disciplinary classification.

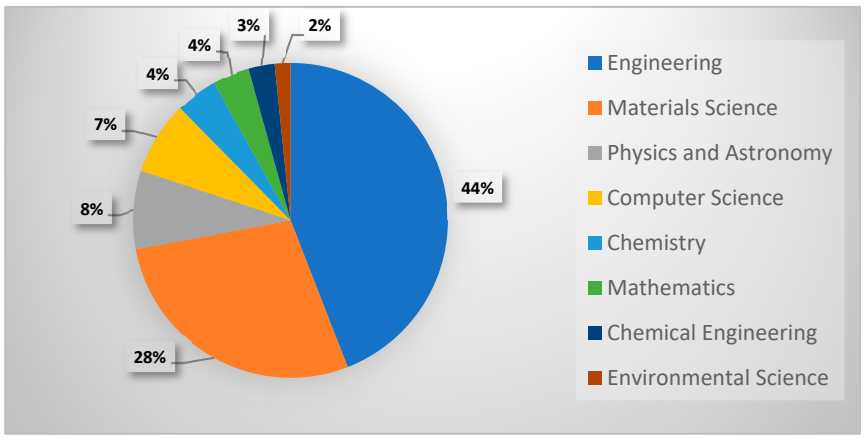

Figure 5. Disciplinary distribution of publications on AI–GFRP research (2006–2025). Data source: Scopus (query in Section 3.3).

The strong representation of engineering underscores the application-driven nature of this research area. Most studies aim to improve the mechanical performance, durability, and structural reliability of GFRP-reinforced systems core issues in civil and structural engineering [57]. Given that GFRP bars are primarily employed as reinforcement elements in concrete structures, their mechanical behavior, bond performance, and failure mechanisms are typically explored through engineering frameworks. Within this context, AI acts as a predictive and optimization tool for enhancing structural design, damage prediction, and safety assessment [58,59].

The significant share of materials science (28%) highlights the growing importance of AI in understanding and optimizing microstructural, thermal, and physico-chemical properties of composite materials. Researchers increasingly employ AI to predict material behavior, optimize fabrication parameters, and accelerate material innovation cycles, signaling a shift toward intelligent materials design and multi-scale characterization [60].

Disciplines such as physics (8%) and computer science (7%) play complementary roles as methodological enablers. Physics contributes through the development of constitutive models and simulation frameworks that describe composite behavior under complex stress states, while computer science provides the machine learning algorithms, neural networks, and data-driven models that enable such predictions [61].

Smaller contributions from chemistry (4%), mathematics (4%), and chemical engineering (3%) are associated with specialized studies on resin chemistry, fibre–matrix interface modeling, and advanced mathematical formulations for composite mechanics. Despite their limited volume, these areas enhance the theoretical and computational foundations of the field.

Finally, the marginal presence of environmental science (2%) reveals that sustainability assessments and life cycle analyses remain underexplored in AI–GFRP research [62]. This underrepresentation points to a critical research gap and an opportunity for future studies, particularly within the context of decarbonization, recycling, and circular economy strategies [63].

Overall, the disciplinary distribution indicates that while the field is still anchored in engineering and materials science, it is progressively evolving toward interdisciplinarity, integrating computational, physical, and environmental perspectives. This trend highlights the growing convergence of digital technologies and sustainable design principles in advancing smart, eco-efficient GFRP-based systems.

4.4. Analysis by Country

Figure 6 illustrates the geographical distribution of publications on the application of Artificial Intelligence (AI) to Glass Fibre Reinforced Polymer (GFRP) bars. The analysis reveals a clear dominance of China (27%) and India (18%), followed by the United States (10%), Saudi Arabia (7%), United Kingdom (6%), Canada (6%), Germany (6%), Iran (5%), South Korea (4%), and Pakistan (3%).

This distribution confirms the global expansion of AI-based GFRP research, with Asia emerging as the most active hub of scientific production.

The leading position of China reflects the country's sustained investments in infrastructure modernization and digital innovation, which have fostered a strong synergy between smart materials and computational design tools [64]. Large-scale national programs on transportation, coastal protection, and green construction have accelerated the use of lightweight, durable, and corrosion-resistant reinforcements such as GFRP bars [65]. Consequently, AI-driven predictive modeling has become a key component of Chinese research in optimizing structural behavior, durability, and life-cycle performance.

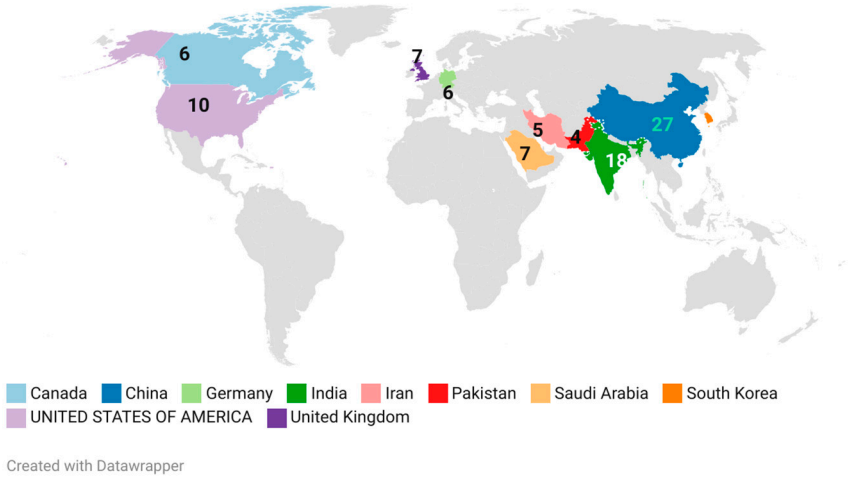

Figure 6. Paper distribution per country (created by the authors using Datawrapper GmbH, Berlin, Germany; https://www.datawrapper.de, accessed on 20 February 2025).

India, ranking second with 18% of publications, mirrors a similar trend of rapid technological adoption. Its strong academic output in civil and structural engineering, combined with government-led initiatives on digital transformation in construction, explains this visibility [66]. Indian researchers increasingly rely on AI to reduce experimental costs, shorten testing cycles, and improve material characterization through data-driven approaches [67].

The United States (10%), together with European contributors such as the United Kingdom (6%) and Germany (6%), represents a mature research bloc shifting from purely experimental work to hybrid AI–mechanistic modeling frameworks [68]. Their long-standing interdisciplinary collaborations between engineering, materials, and computer science laboratories promote the application of machine learning to structural health monitoring, reliability analysis, and durability prediction [69].

The emergence of countries such as Saudi Arabia (7%), Iran (5%), South Korea (4%), and Pakistan (3%) signals a growing regional diversification of research efforts. These nations are integrating AI into construction and materials science under national sustainability and innovation strategies, seeking alternatives to traditional steel reinforcement [70]. Many of these studies are conducted in partnership with international universities, strengthening the global collaborative network.

Overall, the wide geographical spread of contributions from North America and Europe to Asia and the Middle East demonstrates that the integration of AI into GFRP

research has evolved into a globally competitive and interdisciplinary field. This worldwide engagement underscores the strategic relevance of AI-enhanced composites in advancing sustainable, resilient, and intelligent infrastructure systems [66].

4.5. Influence of Authors

Figure 7 presents the co-authorship network generated from the 102 Scopus-indexed publications, highlighting three distinct clusters that define the collaborative landscape of research on Artificial Intelligence (AI) applied to Glass Fibre Reinforced Polymer (GFRP) bars.

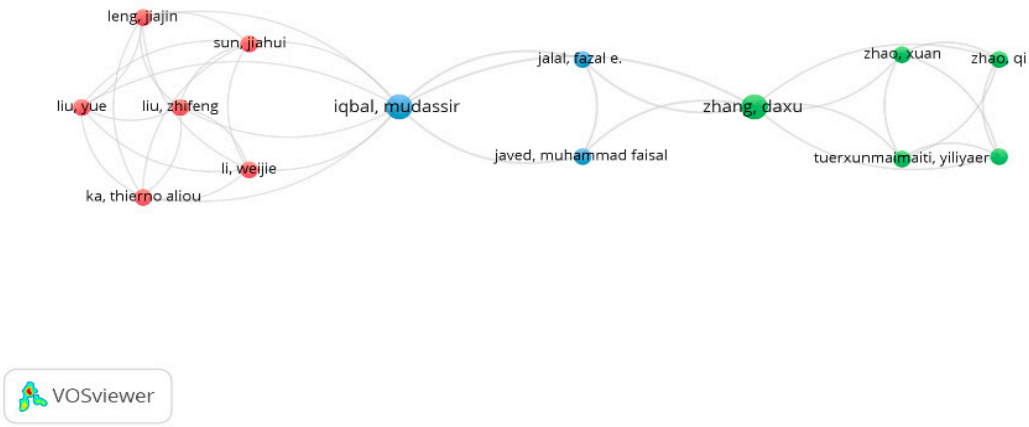

Figure 7. Co-authorship network visualization (VOSviewer 1.6.20) showing three principal clusters of collaboration in AI–GFRP research (2006–2025).

The network structure, produced using fractional counting in VOSviewer (v1.6.20), reveals a moderate level of collaboration, with a limited number of authors forming strong, recurring partnerships. This reflects a field that is still consolidating around a few active research groups rather than a globally integrated network.

The interpretation follows the methodological principles of Van Eck and Waltman (2010) [55], which quantify collaboration intensity based on co-authorship link strength.

- **Cluster 1—Chinese research network (red cluster)**

This cluster, composed of Liu Zhifeng, Liu Yue, Li Weijie, Leng Jiajin, Sun Jiahui, and Ka Thierno Aliou, represents the most cohesive and interconnected collaboration group in the field.

The authors within this cluster are primarily affiliated with Chinese universities and research institutions, focusing on the mechanical characterization, durability, and optimization of GFRP-reinforced systems through machine learning and neural-network approaches [71,72].

Their strong internal connections reflect ongoing institutional collaborations and joint publications, supported by shared datasets and experimental programs [73].

- **Cluster 2—Modeling and simulation network (green cluster)**

The second cluster brings together Zhang Daxu, Zhao Qi, Zhao Xuan, and Tuernxunmaimaiti Yiliyaer.

These researchers concentrate on AI-assisted modeling, numerical simulation, and multi-scale analysis of GFRP materials. Their work often bridges the gap between material-level studies and structural-scale applications, linking computational optimization with practical reinforcement design [74,75].

This cluster acts as a bridge between experimental research (Cluster 1) and international collaborations (Cluster 3), illustrating the emerging interdisciplinarity of AI-based composite materials research.

- **Cluster 3—South Asian collaborative network (blue cluster)**

Cluster 3 consists of Iqbal Mudassir, Jalal Fazal E., and Javed Muhammad Faisal, representing research teams from South Asia.

Their studies emphasize machine-learning-based damage detection, predictive modeling, and structural health monitoring of fiber-reinforced polymer systems [76,77].

Although smaller in size, this cluster demonstrates strong inter-institutional links and contributes to the diversification of global AI–GFRP research, establishing valuable collaborations across developing research regions.

- **Overall interpretation**

The overall co-authorship map demonstrates a field that is in transition from regional concentration toward international integration.

While Chinese authors dominate in terms of publication volume and network cohesion, the growing presence of clusters from South and Central Asia indicates a broadening of the research community.

The network topology suggests a maturing scientific ecosystem, in which AI serves as a common methodological foundation connecting experimental, computational, and structural approaches [78].

Such a configuration marks the evolution of AI–GFRP research toward a more interdisciplinary and globally collaborative domain.

Table 1 presents the top 10 most influential authors based on fractionalized publication counts and total link strength (TLS). These quantitative indicators complement the visual clusters in Figure 7, providing a clearer overview of key contributors to the field.

Table 1. Top 10 most influential authors in AI–GFRP research (VOSviewer data).

Rank	Author	Cluster	Fractionalized Publications	Total Link Strength (TLS)	Citations Within Corpus
1	Zhang, Daxu	2	2	7	127
2	Iqbal, Mudassir	3	2	9	118
3	Jalal, Fazal E.	3	1	3	117
4	Javed, Muhammad Faisal	3	1	3	117
5	Tuerxunmaimaiti, Yiliyaer	2	1	4	10
6	Zhang, Pei Fu	2	1	4	10
7	Zhao, Qi	2	1	4	10
8	Zhao, Xuan	2	1	4	10
9	Li, Weijie	1	1	6	1
10	Liu, Zhifeng	1	1	6	1

The co-authorship analysis (Figure 7) revealed three major clusters of collaboration. Cluster 1 (red) includes Chinese researchers such as Liu Zhifeng, Li Weijie, and Sun Jiahui, who primarily investigate experimental and mechanical aspects of AI-enhanced GFRP composites. Cluster 2 (green), led by Zhang Daxu and Zhao Qi, focuses on modeling and simulation approaches, with Zhang Daxu exhibiting the highest total link strength (TLS = 7). Cluster 3 (blue), composed of Iqbal Mudassir, Jalal Fazal E., and Javed Muhammad Faisal,

reflects emerging international collaborations from South Asia, particularly in damage detection and AI-based structural optimization. Together, these clusters illustrate both the regional consolidation of research in China and the increasing participation of global contributors in AI–GFRP studies.

4.6. Cluster Analysis

Current trends in the integration of Artificial Intelligence (AI) into GFRP bar research were identified using keyword co-occurrence analysis. Cluster visualization (Figure 8) was used to obtain a global view of the intellectual structure of this emerging field. Co-occurrences in the corpus of documents made it possible to capture the underlying patterns, providing insights into how AI is progressively transforming the study and application of fiber-reinforced composites.

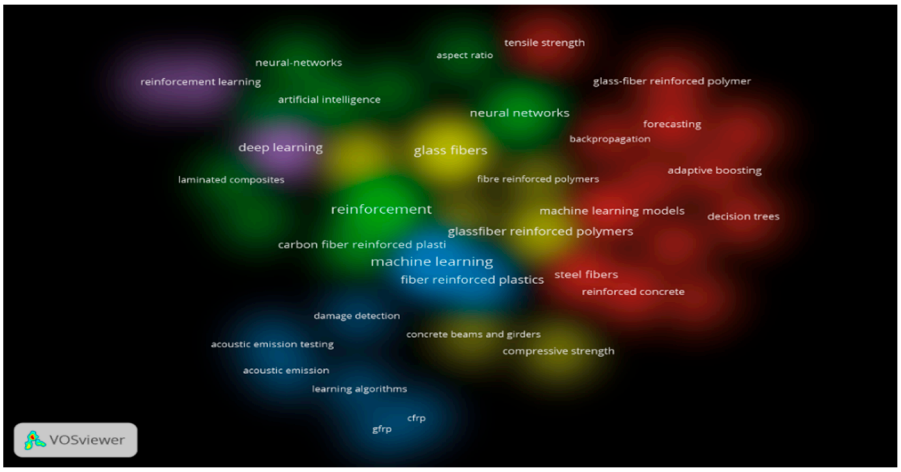

Figure 8. Visualization of cluster density (Source: VOSviewer).

The visual representation of the clusters revealed five distinct groups, each composed of a set of keywords with similar proximity patterns. In VOSviewer, these clusters are identified by numbers and colors. Each point in the visualization is represented by a specific color reflecting the density of the elements. In addition to the default shade assigned by VOSviewer, we have given each cluster a descriptive label to better characterize it. These labels were defined according to the main research focus of each cluster, as shown in Table 2: Machine learning applications in structural analysis (Cluster 1 "red") comprises 13 items. The other clusters have been entitled: AI-driven composite materials modeling (Cluster 2 "green", 10 items), Smart damage detection systems (Cluster 3 "blue", 8 items), Mechanical properties of fiber reinforced materials (Cluster 4 "yellow", 7 items), and Advanced learning systems for glass fiber analysis (Cluster 5 "purple", 3 items).

This classification highlights the rapid evolution of the field, from the early application of machine learning for structural prediction to the development of advanced monitoring systems and deep learning-based optimization approaches. It reflects a transition from basic performance modeling to intelligent, adaptive systems capable of supporting the large-scale adoption of GFRP bars in sustainable construction.

The co-occurrence analysis of keywords generated five distinct thematic clusters (Figure 8), summarized in Table 2. Each cluster represents a specific research orientation, illustrating the progressive integration of Artificial Intelligence (AI) into the study of GFRP bars from structural performance prediction to advanced material modeling and smart monitoring.

Table 2. Authors keywords' co-occurrence network.

Cluster	Color	34 Items	Main Items	Cluster Label
Cluster 1	Red	13	adaptative boosting, backpropagation, decision trees, forecasting, glass-fiber reinforced polymer bars, learning systems, machine learning mode, random forests, reinforced concrete, sensitivity analysis, steel corrosion, steel fibers, tensile strength	Machine Learning Applications in Structural Analysis
Cluster 2	Green	10	artificial intelligence, aspect ratio, carbon fiber reinforced plastics, convolutional neural networks, finite element method; laminated composites, neural networks, neural-networks, reinforcement, thermography (imaging)	AI-Driven Composite Materials Modeling
Cluster 3	Blue	8	acoustic emission, acoustic emission testing, cfrp, damage detection, fiber reinforced plastics, gfrp, learning algorithms, machine learning	Smart Damage Detection Systems
Cluster 4	Yellow	7	compressive strength, concrete beams and girders, fiber reinforced concrete, fiber reinforced polymers, glass fiber reinforced plastics, glass fibers, glassfiber reinforced polymers (gfrp)	Mechanical Properties of Fiber Reinforced Materials
Cluster 5	Purple	3	deep learning, glass fiber, reinforcement learning	Advanced Learning Systems for Glass Fiber Analysis

Cluster 1: "Machine Learning Applications in Structural Analysis"

The bibliometric analysis indicates that this cluster forms the core convergence between machine learning (ML) techniques and structural engineering applications involving GFRP bars. This area has gained significant prominence due to the growing demand for predictive models capable of forecasting the long-term mechanical and durability performance of composite reinforcement systems under diverse loading and environmental conditions [76].

Machine learning algorithms such as adaptive boosting, decision trees, and random forests have been extensively applied to predict key performance parameters, including tensile strength, bond behavior, and durability characteristics of GFRP bars [77]. These AI-driven methods provide substantial advantages over traditional experimental approaches by reducing testing time and costs, while enabling comprehensive sensitivity analyses to identify the most influential design and environmental variables [78].

The strong presence of the keyword "steel corrosion" reflects a comparative research perspective, wherein AI-based models are used to demonstrate the superior corrosion resistance of GFRP relative to conventional steel reinforcement [63,79,80]. This comparative framework has become instrumental in establishing the practical and economic value of GFRP bars, particularly in marine, coastal, and chemically aggressive environments [63,80,81]. Similarly, the frequent co-occurrence of "reinforced concrete" indicates that the focus of these studies extends beyond theoretical modeling to real-world structural applications, emphasizing the integration of AI into concrete structural design and performance assessment [73,74,82].

Despite these advances, methodological challenges persist, particularly concerning the standardization of model validation protocols and the availability of robust, high-quality datasets that accurately capture the full spectrum of GFRP performance scenarios [75]. Variability in manufacturing processes, constituent materials, and testing protocols continues to hinder the development of universally applicable models [35,83].

Future research should focus on the development of ensemble modeling strategies that combine multiple ML algorithms (e.g., boosting, bagging, and stacking) to enhance

prediction accuracy, interpretability, and cross-domain generalization [77,78]. Establishing shared open-access datasets and benchmark validation frameworks will also be critical for ensuring model transparency and reproducibility, thereby facilitating the integration of AI-based predictive modeling into engineering design standards and codes.

Cluster 2: "AI-Driven Composite Materials Modeling"

This cluster represents the integration of Artificial Intelligence (AI) with advanced computational modeling techniques, focusing primarily on the optimization of composite material design and manufacturing processes [37,38].

The convergence of convolutional neural networks (CNNs) with the finite element method (FEM) exemplifies how AI is transforming traditional computational-mechanics approaches and enabling faster, more adaptive simulations of composite behavior under different stress conditions [13,34].

The presence of thermography (imaging) in this cluster highlights AI's growing role in real-time quality control during manufacturing. Computer-vision algorithms coupled with thermal imaging make it possible to automatically detect voids, delaminations, and irregular fiber distributions, ensuring the structural integrity of GFRP bars and other composite products [1,82]. This represents a key step toward automated defect detection and process optimization in industrial composite fabrication.

The integration of laminated composites and carbon-fiber-reinforced plastics (CFRP) within this cluster suggests that techniques initially developed for GFRP are now being generalized to other composite families, supporting the creation of unified AI frameworks for modeling diverse fiber-reinforced materials [84,85].

The keyword "aspect ratio" underscores the relevance of fiber-geometry optimization through AI-driven design. Machine-learning algorithms are increasingly used to determine optimal fiber orientations, lengths, and distributions, thereby enhancing mechanical performance while reducing raw-material use and production costs [37,86].

Despite these advances, major challenges remain in scaling AI-optimized composite designs to industrial production. The complexity of composite manufacturing requires AI models that consider real-world process constraints such as temperature gradients, cure kinetics, and equipment limitations [1,87,88]. Developing robust hybrid frameworks that couple physics-based and data-driven modeling will be essential to bridge the gap between simulation accuracy and manufacturability.

Cluster 3: "Smart Damage Detection Systems"

This cluster highlights the growing integration of Artificial Intelligence (AI) into structural health monitoring (SHM) and damage detection of fiber-reinforced polymer (FRP) systems. The combination of acoustic emission analysis and learning algorithms marks a paradigm shift toward intelligent monitoring capable of providing continuous, real-time assessments of structural integrity [84].

By coupling machine-learning models with acoustic-emission testing, researchers can now automatically classify various damage mechanisms such as fiber rupture, matrix cracking, and delamination [36]. These AI-enhanced systems offer markedly higher sensitivity than traditional non-destructive testing (NDT) approaches, enabling earlier identification of deterioration and more accurate prediction of the remaining service life [8].

The coexistence of both GFRP and CFRP terms within this cluster indicates that such AI-driven diagnostic frameworks are being developed as universal platforms adaptable to different composite types [89]. This versatility is crucial for practical deployment in infrastructures bridges, tunnels, and marine structures where several composite materials may coexist [90].

The use of learning algorithms for damage detection represents a major step toward predictive maintenance strategies [86]. By extracting patterns from acoustic-emission data, AI systems learn to recognize precursor signals that precede failure, allowing timely maintenance actions and reducing the risk of catastrophic structural collapse [91].

Despite these advances, implementation challenges remain, including sensor durability, signal reliability in harsh environments, and the need for standardized, large-scale datasets that encompass diverse loading and environmental conditions [92]. Future work should prioritize self-calibrating and adaptive systems able to maintain accuracy over long-term monitoring and under variable field conditions [93].

These developments strengthen the bridge between data-driven diagnostics and mechanical-property modeling discussed in the next cluster, underscoring the maturity and practical relevance of AI-based SHM in composite-reinforced infrastructures.

Cluster 4: "Mechanical Properties of Fiber Reinforced Materials"

This cluster represents the foundational research base supporting AI applications in GFRP studies, emphasizing the mechanical characterization and structural behavior of fiber-reinforced materials [25,57]. The strong presence of terms such as compressive strength, concrete beams, and girders indicates that these studies primarily address load-bearing applications and the optimization of structural performance [57,58].

The coexistence of multiple terminological variants referring to GFRP materials within this cluster underlines the persistent standardization challenges in the field of composite materials [73,74]. Such terminological inconsistencies often hinder data integration across studies and complicate the creation of unified databases, a critical step for developing reliable AI models. Establishing standardized nomenclature and data-reporting protocols would significantly enhance knowledge synthesis and facilitate automated data-driven learning systems.

Research within this cluster provides the empirical foundation required to train and validate the machine learning algorithms identified in previous clusters [76,77]. The experimental data generated through mechanical testing campaigns form the benchmark datasets for AI-based predictive models designed to estimate GFRP performance under various mechanical and environmental conditions [57,73,82].

The focus on fiber-reinforced concrete applications reflects the practical orientation of this body of work toward real-world structural implementations [57,73,74]. It also demonstrates that AI developments are being guided by the most commercially relevant contexts, such as bridge decks, marine structures, and precast components, where GFRP reinforcement offers tangible durability and weight advantages.

Despite its fundamental contribution, this research domain still faces critical knowledge gaps concerning the long-term degradation of mechanical properties under coupled environmental and mechanical effects [45,57]. Future studies should integrate accelerated aging experiments with AI-driven analytical frameworks to predict durability trends and extend the service life modeling of GFRP-reinforced structures

Cluster 5: "Advanced Learning Systems for Glass Fiber Analysis"

This cluster, although the smallest, represents the most advanced and exploratory frontier of AI applications in glass fiber research [77]. The recurring presence of deep learning and reinforcement learning reflects a growing scientific interest in leveraging high-level AI architectures to optimize glass fiber performance, process efficiency, and material innovation.

Deep learning techniques demonstrate remarkable potential for analyzing complex, high-dimensional datasets that describe the intricate relationships between manufacturing parameters, microstructural configurations, and macroscopic mechanical behavior [77]. By

uncovering non-linear correlations often invisible to traditional statistical or regression-based models, these algorithms enable predictive insights into fiber optimization and structural performance [77].

Reinforcement learning, on the other hand, introduces a paradigm of autonomous process control in composite manufacturing. Through reward-based learning mechanisms, production systems can continuously adjust key processing parameters to maximize performance indicators such as tensile strength or fiber alignment while minimizing energy consumption and material waste [79].

The emphasis on glass fiber rather than the broader GFRP terminology indicates that this cluster focuses on fundamental material-level optimization, including fiber formation, sizing chemistry, and surface modification processes [77]. Such work lays the groundwork for next-generation GFRP systems, providing the data foundations required for the higher-level structural modeling described in earlier clusters.

However, the small size of this cluster suggests that these advanced AI techniques are still in their early adoption phase within the glass fiber community. The computational cost, data scarcity, and lack of open standardized datasets continue to limit their broad deployment [76,78]. Addressing these challenges will require the development of user-friendly, resource-efficient frameworks that democratize access to AI tools and allow researchers to apply deep and reinforcement learning without extensive programming expertise [88].

A temporal analysis of author keywords shows that the introduction of advanced learning techniques in GFRP research is a recent phenomenon. The term "deep learning" first appeared in publications after 2020, with a sharp increase to eight papers between 2022 and 2025. "Reinforcement learning" emerged even more recently, identified in two studies published after 2023, mainly focused on adaptive structural optimization and self-healing control systems. Hybrid or physics-informed approaches began to appear in three publications from 2024 onward, combining data-driven predictions with constitutive or finite-element models to enhance generalization. This chronological pattern indicates a shift from conventional machine-learning models toward more autonomous and physics-aware AI frameworks, reflecting the growing maturity and interdisciplinarity of this field.

Overall, this cluster underscores the technological maturation of the field from traditional mechanical characterization to autonomous, AI-driven material design and manufacturing optimization. It completes the thematic continuum identified across previous clusters, highlighting how AI is progressively transforming the entire life cycle of GFRP materials from fiber synthesis to structural performance monitoring.

4.7. Thematic Evolution and Emerging Trends

The thematic evolution of research on the application of Artificial Intelligence (AI) to Glass Fibre Reinforced Polymer (GFRP) bars demonstrates a clear transition from exploratory computational modeling to integrated smart and sustainable structural systems. The five clusters identified in the bibliometric analysis collectively trace this progression, revealing both the diversification of research aims and the consolidation of methodological frameworks [94,95].

Early studies (Cluster 4—Mechanical Properties of Fiber Reinforced Materials) primarily focused on the experimental characterization of mechanical and durability properties, providing the foundational datasets required for AI-driven modeling [25,57]. As computational capacity increased, machine learning approaches (Cluster 1) emerged as tools for predicting tensile and compressive strength, optimizing mix proportions, and eval-

uating the performance of reinforced concrete elements under various environmental conditions [94,96,97].

Subsequent developments in Cluster 2—AI-Driven Composite Materials Modeling marked a shift toward integrated design and simulation frameworks, where convolutional neural networks and finite element analysis are jointly applied to optimize composite manufacturing and detect production defects through thermographic imaging [20,34,79]. These approaches reflect a growing emphasis on data-informed materials engineering, bridging the gap between material science and structural performance [13,75].

More recent research in Cluster 3—Smart Damage Detection Systems demonstrates a move toward real-time monitoring and predictive maintenance, integrating AI with non-destructive testing methods such as acoustic emission analysis [36,84,89]. This trend illustrates the evolution from static predictive models to dynamic learning systems capable of autonomous decision-making and self-diagnosis [86,91].

Emerging contributions (Cluster 5—Advanced Learning Systems for Glass Fiber Analysis) highlight the early adoption of deep learning and reinforcement learning, signaling the next frontier of intelligent and adaptive design systems [77,79]. These methods aim to achieve autonomous optimization of manufacturing parameters, improved sustainability, and reduced energy consumption [78,88].

Overall, the thematic trajectory indicates a progressive shift from descriptive to prescriptive and self-learning systems, where AI no longer serves solely as an analytical tool but as a core enabler of smart, adaptive, and sustainable composite structures [47,95]. The growing inclusion of interdisciplinary fields such as computer vision, sensor fusion, and environmental modeling suggests that the next stage of evolution will focus on integrating AI with digital twin technologies and life-cycle sustainability assessment frameworks, aligning with the global goals of circular construction and resilient infrastructure [45,76,87].

Recent research has increasingly adopted advanced machine-learning architectures such as Random Forest (RF), Support Vector Machine (SVM), and Convolutional Neural Networks (CNN) for predicting mechanical properties, optimizing mix design, and detecting micro-defects in composite structures [60,67,68,85,86,90]. These algorithms provide enhanced generalization capabilities and can handle heterogeneous datasets common in GFRP material studies.

5. Discussion and Future Research Directions

The findings of this bibliometric analysis highlight a significant scientific convergence between Artificial Intelligence (AI) and composite materials engineering. Over the past decade, research has evolved from performance prediction and computational modeling toward integrated, data-driven frameworks that couple AI with sustainability and smart monitoring. To contextualize these findings, Table 3 provides a comparative overview of this study relative to previous bibliometric reviews addressing AI in composites and in construction materials.

This comparative perspective demonstrates that previous bibliometric reviews have largely focused on either general composites or broad construction processes, without addressing the specific intersection between AI and GFRP materials. The present study fills this gap by integrating both material-level and structural-level perspectives, thereby contributing to a more comprehensive understanding of how AI can advance the design, performance, and sustainability of fiber-reinforced composites.

Table 3. Comparative overview of recent bibliometric studies on AI in composites and construction material.

Study	Scope and Dataset	Main Focus	Analytical Tools/Indicators	Key Findings/ Research Gaps
Bibliometric study on AI in composite materials (Materials Today: Proceedings, 2023)	Focused on AI-assisted modeling of fiber-reinforced composites over the past decade.	Performance prediction, defect detection, and process optimization in composite manufacturing.	Network visualization (VOSviewer) and keyword co-occurrence.	Emphasized mechanical and computational advances; limited attention to sustainability or multi-scale integration.
Bibliometric review on AI in construction engineering (Automation in Construction, 2022)	Covered AI applications in smart construction and project management.	Integration with BIM, robotics, and digital quality control.	Scientometric indicators and cluster analysis.	Highlighted rapid post-2018 growth; lacked material-level insight and cross-domain synthesis.
Present study (AI in GFRP bar development)	Targeted analysis of AI applications in glass-fiber reinforcement systems.	Modeling, optimization, and damage detection in GFRP composites.	Combined quantitative bibliometrics and qualitative thematic synthesis.	First focused review on AI–GFRP synergy; identifies future directions in sustainability, life-cycle modeling, and digital-twin integration.

Research Gaps and Future Research Opportunities

The bibliometric assessment reveals that, despite significant advances in applying Artificial Intelligence (AI) to Glass Fibre Reinforced Polymer (GFRP) research, several key areas remain insufficiently explored or standardized. Addressing these gaps is crucial for transitioning from exploratory research to practical, large-scale implementation.

- **Integration of IoT, digital twins, and real-time monitoring.**

Emerging technologies such as the Internet of Things (IoT) and digital twins remain underutilized in GFRP research. While AI models can predict mechanical and durability performance, few studies incorporate sensor-based, real-time monitoring frameworks that enable adaptive learning and structural feedback. Integrating IoT networks and digital twin simulations could significantly improve predictive maintenance and early damage detection in reinforced structures [18,73,75].

- **Sustainability and life-cycle assessment (LCA).**

The potential of AI to support sustainable material design remains largely unexplored. Most studies focus on mechanical optimization without considering environmental impact. Incorporating life-cycle assessment (LCA) and eco-design parameters into AI frameworks would enable the reduction in the carbon footprint of GFRP materials and foster circular-economy practices. Such integration could promote the design of low-impact composites with enhanced recyclability and resource efficiency [76,78,98].

- **Standardization, validation, and hybrid modeling.**

A persistent challenge is the lack of standardized validation protocols and open-access datasets for benchmarking AI models. Inconsistent data formats and algorithmic procedures often hinder reproducibility and comparison across studies. Establishing universal performance metrics, common datasets, and validation schemes would strengthen the reliability of AI-based research [68,99]. Moreover, hybrid models combining deep learning, finite element analysis (FEA), and traditional regression techniques can improve prediction accuracy while preserving interpretability [77,84].

- **Bridging the gap between theory and practice.**

Despite numerous simulation-based advances, the industrial implementation of AI-driven approaches remains limited. Factors such as data scarcity, sensor degradation, and environmental variability pose significant barriers to field deployment [75,85,86]. The development of real-time adaptive systems capable of autonomous calibration and feedback control represents an essential step toward fully operational smart construction ecosystems.

- **Toward robust and sustainable AI frameworks.**

Future research should prioritize the development of robust and adaptive AI frameworks that can integrate heterogeneous datasets, adapt to dynamic conditions, and learn continuously from real-time information [5]. Moreover, ensuring transparency, interpretability, and sustainability integration within AI systems will enable the creation of efficient and eco-responsible reinforcement materials [73,82,95].

6. Conclusions

This paper delivers the first systematic and reproducible bibliometric mapping of how Artificial Intelligence (AI) is transforming the development, optimization, and assessment of Glass Fibre Reinforced Polymer (GFRP) bars.

By consolidating 102 Scopus-indexed studies published between 2006 and 2025, it quantitatively reveals the evolution, collaboration patterns, and thematic structure of this interdisciplinary field bridging civil engineering, materials science, and computer science.

(1) Scientific Contributions: The analysis confirms an exponential growth in AI–GFRP publications since 2015, with a compound annual growth rate of 10.1%, reflecting the maturation of the domain from exploratory research to a coherent scientific discipline.
Five thematic clusters were identified: (i) machine learning in structural analysis, (ii) AI-driven composite modeling, (iii) smart damage detection, (iv) mechanical characterization, and (v) advanced deep learning frameworks.
The study demonstrates the geographical leadership of China, India, and the United States, while highlighting emerging contributions from Saudi Arabia, Iran, and South Korea, illustrating a shift toward global and collaborative research ecosystem.
The methodological transparency of this study using the PRISMA workflow, inter-rater validation ($\kappa = 0.81$), and an open cleaned dataset provides a replicable framework for future bibliometric analyses in engineering sciences.

(2) Technological and Practical Insights: AI techniques such as neural networks, random forests, and convolutional models are now key enablers for performance prediction, quality control, and design optimization in GFRP manufacturing.
The integration of AI-based inspection systems (acoustic emission, thermography, digital imaging) paves the way for real-time monitoring and predictive maintenance in composite structures.
Emerging hybrid and physics-informed algorithms promise improved generalization and interpretability, addressing one of the main industrial barriers identified by reviewers.

(3) Identified Gaps and Challenges: Sustainability and life-cycle analysis remain underrepresented (≈2% of the corpus). Future work should explicitly connect AI models with environmental and circular-economy indicators.
The absence of standardized validation protocols for AI models limits comparability across studies; the community should prioritize benchmark datasets and cross-laboratory validation frameworks.
Barriers to industrial adoption such as data heterogeneity, privacy constraints, and

lack of regulatory guidance must be overcome through open data initiatives and international standardization efforts.

(4) Future Research Directions

- Develop multi-source bibliometric extensions integrating Web of Science and gray literature to capture hidden industrial contributions.
- Create open-access datasets linking mechanical, chemical, and environmental data for model benchmarking.
- Promote interdisciplinary funding mechanisms that unite engineers, material scientists, and computer scientists.
- Integrate AI and sustainability metrics into design standards to support green and resilient infrastructure.

(5) Overall Impact: This research fills a critical scientific gap by quantitatively mapping the emergence of AI–GFRP integration and offering a clear research agenda for the next decade.

By combining bibliometric transparency, methodological rigor, and sustainability-oriented recommendations, it provides a reference framework for both academics and practitioners seeking to advance smart, safe, and eco-efficient composite reinforcement technologies.

Future progress in this field will rely on strong interdisciplinary collaboration between materials scientists, computer engineers, and structural designers. Such synergies will accelerate the development of standardized datasets, explainable AI frameworks, and sustainable composite design approaches that align with the goals of circular and intelligent construction.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/buildings16030524/s1.

Author Contributions: Conceptualization, H.Z. and O.D.; methodology, H.Z.; software, H.Z.; validation, H.Z., O.D. and I.A.; formal analysis, H.Z.; investigation, H.Z.; resources, O.D.; data curation, H.Z.; writing—original draft preparation, H.Z.; writing—review and editing, H.Z., O.D. and I.A.; visualization, H.Z.; supervision, O.D.; project administration, O.D.; funding acquisition, O.D. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The data presented in this study are available upon request from the corresponding author. The data are not publicly available due to ongoing related research and confidentiality agreements with research partners.

Acknowledgments: The authors acknowledge the use of ChatGPT for generating Figure 1, and DeepL Translator for assisting in the English translation during manuscript preparation.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Talabi, S.I.; Tobin, J.; Strom, B.; Brownstein, I.; Kunc, V.; Hassen, A.A. Recent and future developments in pultrusion technology with consideration for curved geometries: A review. *Compos. Part B Eng.* **2024**, *283*, 111678. [CrossRef]
2. Sacco, C.; Baz Radwan, A.; Anderson, A.; Harik, R.; Gregory, E. Machine learning in composites manufacturing: A case study of Automated Fiber Placement inspection. *Compos. Struct.* **2020**, *250*, 112514. [CrossRef]
3. Yang, J.; Li, S.; Wang, Z.; Dong, H.; Wang, J.; Tang, S. Using deep learning to detect defects in manufacturing: A comprehensive survey and current challenges. *Materials* **2020**, *13*, 5755. [CrossRef] [PubMed]
4. Yang, Y.; Ahmed, K.; Zhang, R.; Liu, R.; Fortin, G.; Hamada, H.; Ma, Y. A study on the energy absorption capacity of braided rod composites. *Compos. Struct.* **2018**, *206*, 933–940. [CrossRef]

5. Badini, S.; Regondi, S.; Pugliese, R. Unleashing the Power of Artificial Intelligence in Materials Design. *Materials* **2023**, *16*, 5927. [CrossRef] [PubMed]
6. Rajak, D.K.; Wagh, P.H.; Linul, E. Manufacturing technologies of carbon/glass fiber-reinforced polymer composites and their properties: A review. *Polymers* **2021**, *13*, 3721. [CrossRef]
7. Dacol, V.; Caetano, E.; Correia, J.R. Comparative study of damping on pultruded gfrp and steel beams. *Polymers* **2021**, *13*, 2201. [CrossRef]
8. Elenchezhian, M.R.P.; Vadlamudi, V.; Raihan, R.; Reifsnider, K.; Reifsnider, E. Artificial intelligence in real-time diagnostics and prognostics of composite materials and its uncertainties—A review. *Smart Mater. Struct.* **2021**, *30*, 083001. [CrossRef]
9. Moustafa, Z.; Asif, M.; Wuni, I.Y. Circular economy in the building sector: A systematic review of environmental, economic, and social dimensions. *Sustain. Futur.* **2025**, *9*, 100690. [CrossRef]
10. Yaghoubi, V.; Kumru, B. Retrosynthetic Life Cycle Assessment: A Short Perspective on the Sustainability of Integrating Thermoplastics and Artificial Intelligence into Composite Systems. *Adv. Sustain. Syst.* **2024**, *8*, 2300543. [CrossRef]
11. Du, S.; Hou, L.; Zhang, G.; Tan, Y.; Mao, P. BIM and IFC Data Readiness for AI Integration in the Construction Industry: A Review Approach. *Buildings* **2024**, *14*, 3305. [CrossRef]
12. Dong, X.; Liu, Y.; Dai, J. Application of Fully Connected Neural Network-Based PyTorch in Concrete Compressive Strength Prediction. *Adv. Civ. Eng.* **2024**, *2024*, 8048645. [CrossRef]
13. Islam, M.Z.; Leo, C.J.; Zou, J.J.; Liyanapathirana, S.; Hu, P. Artificial Intelligence in Civil Engineering Education. In Proceedings of the 3rd International Conference on Advancements in Engineering Education, Sydney, Australia, 20–23 November 2024. [CrossRef]
14. Khan, S. The Application of Artificial Intelligence and Machine Learning in Civil Engineering. *Doupe J. Top Trending Technol.* **2025**, *1*, 1–11. [CrossRef]
15. Wu, C.; Ma, G.; Hwang, H.-J.; Kim, D.-J. Machine learning-based identification of the relationship between bond strength of GFRP bars and AE hits in concrete beams. *Eng. Struct.* **2025**, *329*, 119845. [CrossRef]
16. Mougenot, B.; Doussoulin, J.-P. Conceptual evolution of the bioeconomy: A bibliometric analysis. *Environ. Dev. Sustain.* **2022**, *24*, 1031–1047. [CrossRef] [PubMed]
17. Amiruddin, M.Z.B.; Samsudin, A.; Suhandi, A.; Coştu, B.; Prahani, B.K. Scientific mapping and trend of conceptual change: A bibliometric analysis. *Soc. Sci. Humanit. Open* **2025**, *11*, 101208. [CrossRef]
18. Pham, X.-L.; Le, T.T. Bibliometric Analysis and Systematic Review of Research on Expert Finding: A PRISMA-guided Approach. *Int. Arab J. Inf. Technol.* **2024**, *21*, 661–674. [CrossRef]
19. Sariyer, G.; Mangla, S.K.; Kazancoglu, Y.; Jain, V.; Ataman, M.G. Data-driven decision making for modelling COVID-19 and its implications: A cross-country study. *Technol. Forecast. Soc. Change* **2023**, *197*, 122886. [CrossRef]
20. De Oliveira, O.J.; da Silva, F.F.; Juliani, F.; Barbosa, L.C.F.M.; Nunhes, T.V.; de Oliveira, O.J.; da Silva, F.F.; Juliani, F.; Barbosa, L.C.F.M.; Nunhes, T.V. Bibliometric Method for Mapping the State-of-the-Art and Identifying Research Gaps and Trends in Literature: An Essential Instrument to Support the Development of Scientific Projects. In *Scientometrics Recent Advances*; IntechOpen: London, UK, 2019. [CrossRef]
21. Thelwall, M.; Sud, P. Scopus 1900–2020: Growth in articles, abstracts, countries, fields, and journals. *Quant. Sci. Stud.* **2022**, *3*, 37–50. [CrossRef]
22. Nogueira, J.P.; da Souza, I.H.S.; Andrade, J.K.S.; Narain, N. Status of research on lactones used as aroma: A bibliometric review. *Food Biosci.* **2022**, *50*, 102004. [CrossRef]
23. Pranckutė, R. Web of Science (WoS) and Scopus: The Titans of Bibliographic Information in Today's Academic World. *Publications* **2021**, *9*, 12. [CrossRef]
24. Azad, S.; Shafei, B. Performance Evaluation of Glass Fiber-Reinforced Polymer (GFRP) Bars in Bridge Decks. In *The 1st International Conference on Net-Zero Built Environment*; Kioumarsi, M., Shafei, B., Eds.; Lecture Notes in Civil Engineering; Springer Nature Switzerland: Cham, Switzerland, 2025; Volume 237, pp. 1361–1370. [CrossRef]
25. Nkurunziza, G.; Debaiky, A.; Cousin, P.; Benmokrane, B. Durability of GFRP bars: A critical review of the literature. *Prog. Struct. Eng. Mater.* **2005**, *7*, 194–209. [CrossRef]
26. Boscato, G.; Ientile, S. Experimental and numerical investigation on dynamic properties of thin-walled GFRP buckled columns. *Compos. Struct.* **2018**, *189*, 273–285. [CrossRef]
27. Chen, Z.; Yu, J.; Nong, Y.; Yang, Y.; Zhang, H.; Tang, Y. Beyond time: Enhancing corrosion resistance of geopolymer concrete and BFRP bars in seawater. *Compos. Struct.* **2023**, *322*, 117439. [CrossRef]
28. Devaraj, R.; Olofinjana, A.; Gerber, C. On the Factors That Determine the Bond Behaviour of GFRP Bars to Concrete: An Experimental Investigation. *Buildings* **2023**, *13*, 2896. [CrossRef]
29. Huang, H.; Yuan, Y.; Zhang, W.; Hao, R.; Zeng, J. Bond properties between GFRP bars and hybrid fiber-reinforced concrete containing three types of artificial fibers. *Constr. Build. Mater.* **2020**, *250*, 118857. [CrossRef]

30. Kou, H.-L.; Li, W.; Zhang, W.-C.; Zhou, Y.; Zhou, X.-L. Stress monitoring on GFRP anchors based on fiber bragg grating sensors. *Sens. Switz.* **2019**, *19*, 1507. [CrossRef] [PubMed]
31. Xu, K.; Ren, C.; Deng, Q.; Jin, Q.; Chen, X. Real-time monitoring of bond slip between GFRP bar and concrete structure using piezoceramic transducer-enabled active sensing. *Sens. Switz.* **2018**, *18*, 2653. [CrossRef] [PubMed]
32. Sharifianjazi, F.; Zeydi, P.; Bazli, M.; Esmaeilkhanian, A.; Rahmani, R.; Bazli, L.; Khaksar, S. Fibre-Reinforced Polymer Reinforced Concrete Members under Elevated Temperatures: A Review on Structural Performance. *Polymers* **2022**, *14*, 472. [CrossRef]
33. Bolina, F.L.; Silva, D.B.; Pachla, E.C. The fire performance of GFRP-RC beams based on FE thermal analysis. *J. Build. Pathol. Rehabil.* **2024**, *10*, 17. [CrossRef]
34. Hussain, A.; Sakhaei, A.H.; Shafiee, M. Machine learning-based constitutive modelling for material non-linearity: A review. *Mech. Adv. Mater. Struct.* **2024**, 1–19. [CrossRef]
35. Zhang, K.; Zhang, K.; Bao, R. Machine learning models to predict the residual tensile strength of glass fiber reinforced polymer bars in strong alkaline environments: A comparative study. *J. Build. Eng.* **2023**, *73*, 106817. [CrossRef]
36. Ma, G.; Du, Q. Optimization on the intellectual monitoring system for structures based on acoustic emission and data mining. *Measurement* **2020**, *163*, 107937. [CrossRef]
37. Chen, Y.; Zhang, J.; Li, Z.; Zhang, H.; Chen, J.; Yang, W.; Yu, T.; Liu, W.; Li, Y. Intelligent methods for optimization design of lightweight fiber-reinforced composite structures: A review and the-state-of-the-art. *Front. Mater.* **2023**, *10*, 1125328. [CrossRef]
38. Nyokum, T.; Tamut, Y. Artificial intelligence in civil engineering: Emerging applications and opportunities. *Front. Built Environ.* **2025**, *11*, 1622873. [CrossRef]
39. Vergara, D.; Lampropoulos, G.; Fernández-Arias, P.; Antón-Sancho, Á. Artificial Intelligence Reinventing Materials Engineering: A Bibliometric Review. *Appl. Sci. Switz.* **2024**, *14*, 8143. [CrossRef]
40. Arumugam, S.; Ravichandran, P.T. Artificial Intelligence in the Construction Industry: A Status Update, Prospects, and Potential Application and Challenges. In *Recent Advances in Civil Engineering: Select Proceedings of ICC-IDEA 2023; Lecture Notes in Civil Engineering*; Springer: Singapore, 2024; pp. 49–57. [CrossRef]
41. Davim, J.P. Composite materials: A bibliometric analysis. *AIMS Mater. Sci.* **2024**, *11*, 1145–1148. [CrossRef]
42. Singh, V.K.; Singh, P.; Karmakar, M.; Leta, J.; Mayr, P. The journal coverage of Web of Science, Scopus and Dimensions: A comparative analysis. *Scientometrics* **2021**, *126*, 5113–5142. [CrossRef]
43. Van Eck, N.; Waltman, L. *VOSviewer Manual: Manual for VOSviewer Version 1.6.20*; Leiden University: Leiden, The Netherlands, 2023.
44. van Eck, N.J.; Waltman, L. Software survey: VOSviewer, a computer program for bibliometric mapping. *Scientometrics* **2010**, *84*, 523–538. [CrossRef] [PubMed]
45. Xie, F.; Tian, W.; Li, S.; Diez, P.; Zlotnik, S.; Gonzalez, A.G. Experimental Study on the Structural Performance of Glass-Fiber-Reinforced Concrete Slabs Reinforced with Glass-Fiber-Reinforced Polymer (GFRP) Bars: A Sustainable Alternative to Steel in Challenging Environments. *Polymers* **2025**, *17*, 1068. [CrossRef]
46. Belciu, A.-C. Performance in Accessing Funding Through Digital Europe Programme. *East. Eur. J. Reg. Stud. EEJRS* **2025**, *11*, 45–62. [CrossRef]
47. Ho, Y.-S. A bibliometric analysis of highly cited articles in materials science. *Curr. Sci.* **2014**, *107*, 1565–1572.
48. Shamir, L. The effect of conference proceedings on the scholarly communication in Computer Science and Engineering. *Sch. Res. Commun.* **2010**, *1*, 020302. [CrossRef]
49. Guide for Authors—Journal of Building Engineering—ISSN 2352-7102. ScienceDirect.com by Elsevier. Available online: https://www.sciencedirect.com/journal/journal-of-building-engineering/publish/guide-for-authors (accessed on 17 September 2025).
50. Guide for Authors—Construction and Building Materials—ISSN 0950-0618. ScienceDirect.com by Elsevier. Available online: https://www.sciencedirect.com/journal/construction-and-building-materials/publish/guide-for-authors (accessed on 17 September 2025).
51. Rane, N.; Desai, P.; Rane, J. Acceptance and integration of Artificial intelligence and machine learning in the construction industry: Factors, current trends, and challenges. In *Trustworthy Artificial Intelligence in Industry and Society*; Deep Science Publishing: San Francisco, CA, USA, 2024; pp. 134–155. [CrossRef]
52. Fazeli-Varzaneh, M.; Ghaderi-Azad, E.; Elango, B. Middle East Countries' Contribution to Global Engineering Research: A Bibliometric Analysis. *Int. J. Inf. Sci. Manag.* **2021**, *19*, 1–13.
53. Naik, C.; Sugimoto, C.; Larivière, V.; Leng, C.; Guo, W. Impact of Geographic Diversity on Citation of Collaborative Research. *Quant. Sci. Stud.* **2023**, *4*, 442–465. [CrossRef]
54. Hao, L.; Yunwei, C.; Biao, Z. Study on the Differences Between Journal Papers and Conference Papers in the Frontier of Basic Research: Taking the Terahertz Field as an Example. In Proceedings of the 20th International Conference on Scientometrics & Informetrics, Yerevan, Armenia, 23–27 June 2025. [CrossRef]

55. Waltman, L.; van Eck, N.J.; Noyons, E.C.M. A unified approach to mapping and clustering of bibliometric networks. *J. Informetr.* **2010**, *4*, 629–635. [CrossRef]
56. Zupic, I.; Čater, T. Bibliometric methods in management and organization. *Organ. Res. Methods* **2015**, *18*, 429–472. [CrossRef]
57. Attia, M.M.; El-Latief, A.A.; Eita, M.A. Performance of RC beams with novelty GFRP under the bending load: An experimental and FE study. *Case Stud. Constr. Mater.* **2023**, *18*, e02000. [CrossRef]
58. Nouri, Y.; Ghanizadeh, A.R.; Safi Jahanshahi, F.; Fakharian, P. Data-driven prediction of axial compression capacity of GFRP-reinforced concrete column using soft computing methods. *J. Build. Eng.* **2025**, *101*, 111831. [CrossRef]
59. Karimipour, A.; Mohebbi Najm Abad, J.; Fasihihour, N. Predicting the load-carrying capacity of GFRP-reinforced concrete columns using ANN and evolutionary strategy. *Compos. Struct.* **2021**, *275*, 114470. [CrossRef]
60. Sorour, S.S.; Saleh, C.A.; Shazly, M. A review on machine learning implementation for predicting and optimizing the mechanical behaviour of laminated fiber-reinforced polymer composites. *Heliyon* **2024**, *10*, e33681. [CrossRef]
61. Huang, H.; Mojumder, S.; Suarez, D.; Amin, A.A.; Fleming, M.; Liu, W.K. Knowledge database creation for design of polymer matrix composite. *Comput. Mater. Sci.* **2022**, *214*, 111703. [CrossRef]
62. Ramachandran, K.; Gnanasagaran, C.L.; Vekariya, A. Life cycle assessment of carbon fiber and bio-fiber composites prepared via vacuum bagging technique. *J. Manuf. Process.* **2023**, *89*, 124–131. [CrossRef]
63. Ascione, F.; Maselli, G.; Nesticò, A. Sustainable materials selection in industrial construction: A life-cycle based approach to compare the economic and structural performances of glass fibre reinforced polymer (GFRP) and steel. *J. Clean. Prod.* **2024**, *475*, 143641. [CrossRef]
64. Yan, J.-K.; Zheng, Z.; Zhou, Y.-C.; Lin, J.-R.; Deng, Y.-C.; Lu, X.-Z. Recent Research Progress in Intelligent Construction: A Comparison between China and Developed Countries. *Buildings* **2023**, *13*, 1329. [CrossRef]
65. Liu, T.; Tang, J.-T.; Zhang, S.; Dong, L.; Hu, L.; Meng, X.; Zhao, Y.; Feng, P. Carbon emissions of durable FRP composite structures in civil engineering. *Eng. Struct.* **2024**, *315*, 118482. [CrossRef]
66. Shruthi, S.; Judson, L.; Paul, V. Addressing the Challenges in Cost Management of Construction Projects by Using AI. *J. Struct. Technol.* **2025**, *10*, 28–47.
67. Luo, D.; Wang, K.; Wang, D.; Sharma, A.; Li, W.; Choi, I.H. Artificial intelligence in the design, optimization, and performance prediction of concrete materials: A comprehensive review. *npj Mater. Sustain.* **2025**, *3*, 14. [CrossRef]
68. Gangwar, A.; Kumar, V.; Yaylaci, M.; Panda, S.K. Computational Modelling and Mechanical Characteristics of Polymeric Hybrid Composite Materials: An Extensive Review. *Arch. Comput. Methods Eng.* **2024**, *31*, 3901–3921. [CrossRef]
69. Sargiotis, D. Advancing Civil Engineering with AI and Machine Learning: From Structural Health to Sustainable Development. *Int. J. Archit. Plan.* **2024**, *4*, 54–81. [CrossRef]
70. Waqar, A.; Othman, I.; Hamah Sor, N.; Alshehri, A.M.; Almujibah, H.R.; Alotaibi, B.S.; Abuhussain, M.A.; Bageis, A.S.; Althoey, F.; Hayat, S.; et al. Modeling relation among implementing AI-based drones and sustainable construction project success. *Front. Built Environ.* **2023**, *9*, 1208807. [CrossRef]
71. Liu, J.; Guo, T.; Wang, L.; Zhang, Y.; Li, J. Assessment model of cyclic compression behaviors for FRP wrapped high-strength concrete-filled steel tubes after seawater exposure. *Eng. Struct.* **2025**, *340*, 120736. [CrossRef]
72. Anjum, N.; Suma; Rakesh, M.; Anjum, R.; Chikkadevegowda, S.; Sandeep, B. The Role of Artificial Intelligence in Microstructural Characterization of Ferrous Materials: A Review of Tools, Findings, and Research Challenges. *J. Mines Met. Fuels* **2025**, *73*, 759–769. [CrossRef]
73. Hadhood, A.; Mohamed, H.M.; Benmokrane, B.; Nanni, A.; Shield, C.K. Assessment of design guidelines of concrete columns reinforced with glass fiber-reinforced polymer bars. *ACI Struct. J.* **2019**, *116*, 193–207. [CrossRef]
74. Guérin, M.; Mohamed, H.M.; Benmokrane, B.; Shield, C.K.; Nanni, A. Effect of glass fiber-reinforced polymer reinforcement ratio on axial-flexural strength of reinforced concrete columns. *ACI Struct. J.* **2018**, *115*, 1049–1061. [CrossRef]
75. Liu, P.; Xia, H. Structure and evolution of co-authorship network in an interdisciplinary research field. *Scientometrics* **2015**, *103*, 101–134. [CrossRef]
76. Machello, C.; Bazli, M.; Rajabipour, A.; Rad, H.M.; Arashpour, M.; Hadigheh, A. Using machine learning to predict the long-term performance of fibre-reinforced polymer structures: A state-of-the-art review. *Constr. Build. Mater.* **2023**, *408*, 133692. [CrossRef]
77. Tian, L.; Wang, L.; Xian, G. Machine learning prediction of residual tensile strength of glass fiber reinforced polymer bars in alkaline environments. *Structures* **2025**, *74*, 108552. [CrossRef]
78. Naveed, M.; Hameed, A.; Rasool, A.M.; Mukhtar, D.; Ahmed, T. AI-based non-linear models for mechanical and toughness properties of sustainable fiber-reinforced geopolymer concrete (FRGPC). *Mech. Adv. Mater. Struct.* **2025**, *32*, 3179–3203. [CrossRef]
79. Rajender, A.; Samanta, A.K.; Paral, A. Comparative study of corrosion-based service life prediction of reinforced concrete structures using traditional and machine learning approach. *Int. J. Struct. Integr.* **2025**, *16*, 591–621. [CrossRef]
80. Alsayed, S.H.; Al-Salloum, Y.A.; Almusallam, T.H. Performance of glass fiber reinforced plastic bars as a reinforcing material for concrete structures. *Compos. Part B Eng.* **2000**, *31*, 555–567. [CrossRef]

81. Zhou, J.-K.; Hao, Z.-H.; Zeng, J.-J.; Feng, S.-Z.; Liang, Q.-J.; Zhao, B.; Feng, R.; Zhuge, Y. Durability assessment of GFRP bars embedded in UHP-ECCs subjected to an accelerated aging environment with sustained loading. *Constr. Build. Mater.* **2024**, *419*, 135364. [CrossRef]
82. Yan, F.; Lin, Z.; Wang, X.; Azarmi, F.; Sobolev, K. Evaluation and prediction of bond strength of GFRP-bar reinforced concrete using artificial neural network optimized with genetic algorithm. *Compos. Struct.* **2017**, *161*, 441–452. [CrossRef]
83. Iqbal, M.; Zhang, D.; Jalal, F.E.; Faisal Javed, M. Computational AI prediction models for residual tensile strength of GFRP bars aged in the alkaline concrete environment. *Ocean Eng.* **2021**, *232*, 109134. [CrossRef]
84. Ghadarah, N.; Ayre, D. A Review on Acoustic Emission Testing for Structural Health Monitoring of Polymer-Based Composites. *Sensors* **2023**, *23*, 6945. [CrossRef] [PubMed]
85. Chen, C.-T.; Gu, G.X. Machine learning for composite materials. *MRS Commun.* **2019**, *9*, 556–566. [CrossRef]
86. Wang, Y.; Soutis, C.; Ando, D.; Sutou, Y.; Narita, F. Application of deep neural network learning in composites design. *Eur. J. Mater.* **2022**, *2*, 118–171. [CrossRef]
87. Rahman, H.; D'Cruze, R.S.; Ahmed, M.U.; Sohlberg, R.; Sakao, T.; Funk, P. Artificial Intelligence-Based Life Cycle Engineering in Industrial Production: A Systematic Literature Review. *IEEE Access* **2022**, *10*, 133001–133015. [CrossRef]
88. Sudheer Kumar Varma, N.; Rajasekhar, P.; Ganesan, G.; Sita Rama Raju, K. Prediction of nano metal matrix composites based on hybrid approach. *Lubr. Sci.* **2024**, *36*, 370–386. [CrossRef]
89. Altin Karataş, M.; Gökkaya, H. A review on machinability of carbon fiber reinforced polymer (CFRP) and glass fiber reinforced polymer (GFRP) composite materials. *Def. Technol.* **2018**, *14*, 318–326. [CrossRef]
90. Dong, W.; Huang, Y.; Lehane, B.; Ma, G. An artificial intelligence-based conductivity prediction and feature analysis of carbon fiber reinforced cementitious composite for non-destructive structural health monitoring. *Eng. Struct.* **2022**, *266*, 114578. [CrossRef]
91. Cardillo, K.A. Artificial Intelligence Approaches for Structural Health Monitoring of Aerospace Structures. Master's Thesis, University of South Carolina, Columbia, SC, USA, 2020.
92. Plevris, V.; Papazafeiropoulos, G. AI in Structural Health Monitoring for Infrastructure Maintenance and Safety. *Infrastructures* **2024**, *9*, 225. [CrossRef]
93. Yang, Y.; Chen, T.; Lin, W.; Jing, M.; Xu, W. Research progress on calibration of bridge structural health monitoring sensing system. *Adv. Bridge Eng.* **2024**, *5*, 32. [CrossRef]
94. Zhang, G.; Chen, C.; Zhang, Y.; Zhao, H.; Wang, Y.; Wang, X. Optimised neural network prediction of interface bond strength for GFRP tendon reinforced cemented soil. *Geomech. Eng.* **2022**, *28*, 599–611. [CrossRef]
95. Ashok, G.; Kumar, P.; Prabhu, T.R. Global Developments in Additive Manufacturing of Polymer Composite Materials: A Scientometric Review. *Arch. Comput. Methods Eng.* **2025**, 1–22. [CrossRef]
96. Zhang, G.; Chen, C.; Li, K.; Xiao, F.; Sun, J.; Wang, Y.; Wang, X. Multi-objective optimisation design for GFRP tendon reinforced cemented soil. *Constr. Build. Mater.* **2022**, *320*, 126297. [CrossRef]
97. Yue, Z.; Ouyang, K.; Yao, X.; Hu, K.; Li, L. A critical review of fiber reinforced polymer bars: A scientometric and visualization analysis. *Clean. Mater.* **2025**, *17*, 100325. [CrossRef]
98. Wang, Y.; Wang, K.; Zhang, C. Applications of artificial intelligence/machine learning to high-performance composites. *Compos. Part B Eng.* **2024**, *285*, 111740. [CrossRef]
99. Chien, C.-F.; Dauzère-Pérès, S.; Huh, W.T.; Jang, Y.J.; Morrison, J.R. Artificial intelligence in manufacturing and logistics systems: Algorithms, applications, and case studies. *Int. J. Prod. Res.* **2020**, *58*, 2730–2731. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.